

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Compute and analyze the correlation between three audio tracks	
Experiment: 7	Date:	Enrollment No: 92400133030

Aim : Image Processing

Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

image_path = "/photo-1728998888313-9bc4eff8e23a.avif"
src_image = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)

if src_image is None:
    print("Error: Image not found!")
else:
    # 1. Image Negation
    negative_image = 255 - src_image

    # 2. Thresholding
    _, thresholded_image = cv2.threshold(
        src_image, 50, 255, cv2.THRESH_BINARY
    )

    # 3. Gamma Correction
    gamma = 2.0

    # Normalize image to [0, 1]
    normalized_image = src_image / 255.0

    # Apply gamma correction
    gamma_corrected_image = np.power(normalized_image, 1 / gamma)

    # Convert back to [0, 255]
    gamma_corrected_image = np.uint8(gamma_corrected_image * 255)

    # Display Images
    plt.figure(figsize=(16, 5))

    # Original
    plt.subplot(1, 4, 1)
    plt.imshow(src_image, cmap="gray")
    plt.title("Original Image")
    plt.axis("off")

    # Negative
    plt.subplot(1, 4, 2)
    plt.imshow(negative_image, cmap="gray")
    plt.title("Negative Image")
    plt.axis("off")

    # Threshold
```

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```
plt.subplot(1, 4, 3)
plt.imshow(thresholded_image, cmap="gray")
plt.title("Threshold Image")
plt.axis("off")

# Gamma
plt.subplot(1, 4, 4)
plt.imshow(gamma_corrected_image, cmap="gray")
plt.title("Gamma Corrected")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Output:

